

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

SKILLS

Languages: English (C1 [IELTS 8.0](#) )
Programming: Python, SQL, Bash, Julia, C/C++, R
Tools: Linux, Git, Docker, pandas, xarray, PyTorch, sklearn, opencv, open3d, nerfstudio, Quarto, $\text{\LaTeX}$
Data Engineering: Large-scale data processing, multi-threading, relational databases (MySQL/PostgreSQL), file formats (GeoTIFF, HDF5, Parquet)

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: [Márton Vaitkus](#) 
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 (**rank 3/34**)
Supervisor: [Galtbayar Artbazar](#) 
Thesis: [Evaluating land degradation using image processing methods](#) 

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- **Data Processing & Python:** Developed custom [Python pipelines](#)  using **xarray** and multi-threading to process large-scale datasets (1,800+ multispectral images). Optimized data ingestion and processing workflows, reducing execution time by 80% (5x factor).
- **Data Infrastructure:** Orchestrated photogrammetry pipelines using **OpenDroneMap** and **Docker** with **GPU acceleration** on Linux. Managed complex dependency environments and GPU debugging to ensure reliable high-throughout processing of spatial data.
- **Data Management:** Utilized **SQL** and relational database principles to organize research metadata and spatial results, facilitating efficient querying and data integrity across multiple field studies.

PROJECTS

[aiSim Data Integration for 3D Gaussian Splatting](#)

- Developed a sensor fusion pipeline to parse and convert over sensor measurements from aiMotive's [aiSim](#)  ADAS simulation platform into the [nerfstudio](#)  3D scene representation format. Worked on the camera coordinate system transformation from sensor frame to the world frame.

[Exploratory Data Analysis – Bike Sharing Dataset](#)

- Performed exploratory data analysis on bike sharing dataset. Published the results personal blog website using Quarto.

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced polynomial regression experiments from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks to using interactive sliders.

AWARDS

Stipendium Hungaricum Scholarship

Sep, 2025